

LITTLE SWAN BALLET ACADEMY

Academy Management System

BRD

Business Requirements Document

Document Type	Business Requirements Document
Version	v3.0 Final
Date	April 2026
Project Lead	Mazen Magdy
Status	Ready for Development
Branch	Miami Branch — Hall 1 & Hall 2
Language	Bilingual — Arabic & English

1. Project Overview

Field	Details
System Name	Little Swan Ballet Academy Management System
الاسم بالعربي	نظام إدارة أكاديمية ليتل سوان للباليه
Branch	Miami Branch (Single Branch — Phase 1)
Halls	Hall 1 & Hall 2 (within Miami branch)
Target Students	Girls aged 3 years and above
Platform	Web Application — Bubble.io (Phase 1)
Language	Bilingual: Arabic & English

1.1 Problem Statement (Current Situation)

- All operations managed manually via Google Sheets and paper notebooks
- Student data duplicated daily — same student re-entered every session
- Attendance, subscriptions, and payments are disconnected — no automation
- Coach hours tracked manually — payroll calculations error-prone
- Inventory (leotards) managed informally — no stock visibility
- No automated alerts for subscription expiry or payment due
- No transfer or exam tracking between levels
- Two halls in the same branch with no conflict prevention

1.2 Project Objectives

- Eliminate manual Excel/paper workflows entirely
- Automate subscription tracking — alert when payment is required
- Enable accurate coach check-in/out and automatic payroll calculation
- Provide real-time daily revenue and expense tracking
- Support student exam results and automatic level promotion
- Manage leotard inventory and custom-order tracking
- Provide clean, role-based access for all staff and coaches
- Prevent scheduling conflicts between Hall 1 and Hall 2
- Support Arabic and English languages throughout the interface

2. Project Scope

2.1 In Scope — Version 1 (MVP)

- Student management — full profile, status, ballet level
- Classes and fixed weekly schedules (Hall 1 or Hall 2 assignment per class)
- Daily session tracking and attendance
- Session-based subscription management (4 sessions per cycle — configurable)
- Payment recording — Cash and InstaPay, including partial payments
- Coach login, check-in, check-out, and automatic payroll calculation
- Student freeze and reactivation (no subscription extension)
- Make-up session assignment (within same subscription period only)
- Student class transfers (staff request → admin approval)
- Exam management and automatic level promotion
- Daily revenue and expense tracking per branch
- Leotard inventory — ready stock and made-to-order tracking
- Events and activities management (recitals, TV shows, workshops)
- Private session tracking and payment logging
- Admin settings panel — subscription plans, levels, exam fees, freeze rules
- Bilingual interface — Arabic and English

2.2 Out of Scope — Version 1

- Student or parent login portal (planned for Phase 2)
- Online payment integration / Paymob / Fawry (Phase 2)
- WhatsApp or SMS automated reminders (Phase 2)
- Mobile application — iOS or Android (Phase 3)
- Second branch management (Phase 2+)

3. User Roles & Permissions

Role	العربي	Count	Access Level
Super Admin	سوبر أدمن	1	Full system — settings, reports, all data
Admin (Technical Director)	المدير الفني / صاحبة المكان	1	Full operational — approve transfers, manage all
Staff (Employee)	موظف	3	Daily ops — attendance, payments, bookings
Coach	مدربة	Variable	Own classes — check-in/out, confirm attendance

3.1 Super Admin Permissions

- Manage all user accounts (create, edit, deactivate)
- Configure system settings: subscription plans, exam fees, level order, freeze limits, absence auto-hide threshold
- View all reports and full financial data across halls
- Override any operation in the system
- Manage hall configuration (Hall 1 and Hall 2)

3.2 Staff (Employee) Permissions

- Add, edit, and manage student records
- Mark attendance for all sessions (present, absent, make-up)
- Record all payment types (subscription, product, event, private, exam)
- Assign make-up sessions within subscription period
- Request student class transfers (pending admin approval)
- Manage leotard sales and custom orders
- Register new student enrollments
- Record daily expenses

3.3 Coach Permissions

- Login to personal account
- View assigned classes and weekly timetable
- Check-in to session at start time
- Check-out from session at end time
- Confirm student attendance (cross-validated against staff entry)

4. Business Requirements & Features

4.1 Student Management

The system maintains a complete, permanent profile for each student. No student record is ever deleted.

Field	Details
Full Name (AR + EN)	Bilingual entry required
Date of Birth	Auto-calculates age dynamically — no manual age entry
Parent Phone	Primary contact number
Ballet Level	Assigned by Admin — linked to Levels configuration
Status	Active / Frozen / Inactive
Notes	Medical, behavioral, or administrative notes
Enrollment Date	Auto-recorded on account creation

Student Status Rules

Status	Behavior	Data
Active	Appears in all sessions, attendance tracked, subscription runs normally	Fully visible
Frozen	Hidden from all sessions. No attendance. No subscription deduction. No make-up allowed.	Preserved — nothing deleted
Inactive	Student left the academy. Hidden from all operations. Can be reactivated if she returns.	Fully preserved permanently

Auto-hide rule: Super Admin configures a maximum consecutive absence count. When exceeded, the system automatically changes the student's status to Inactive.

4.2 Hall & Class Management

The Miami branch has two halls: Hall 1 and Hall 2. Each class is assigned to a specific hall. The system prevents scheduling conflicts between classes in the same hall at the same time.

- Admin creates fixed weekly class slots (not daily — sessions are auto-generated)
- Each class has: Name, Level, Age Group, Default Coach, Hall (1 or 2), Day(s), Start Time, End Time, Max Capacity
- A student can be enrolled in more than one class (e.g., Saturday + Monday groups)
- Hall assignment is enforced — system blocks overlap conflicts

4.3 Daily Session Management

A Session is a single occurrence of a class on a specific date. It is the core daily operational record.

- Auto-generated from the fixed class schedule
- Records: Date, Hall, Responsible Staff, Actual Coach (may differ from default)

- Links to: all attendance records, coach check-in/out, revenue for that day
- Coach can be substituted per session without changing the class default

4.4 Attendance Tracking

Status	Effect on Subscription	Make-up Eligible
Present	-1 session from subscription	No
Absent	No deduction	Yes — within same subscription period
Make-up	-1 session (counts as attended)	No (already a make-up)
Frozen	No entry created — student hidden	No

4.5 Subscription & Make-up Logic

Subscription type: Session-based. Default = 4 sessions per cycle (configurable by Super Admin).

- Each Present or Make-up attendance deducts 1 session from the active subscription
- When remaining sessions = 0 → system shows 'Payment Required' alert next to student name
- Absence does NOT delay payment — cycle is session-count driven, not calendar-driven
- Payment creates a new subscription cycle (new 4-session block starts from payment date)

Make-up Rules

- Make-up is only allowed within the SAME subscription period
- If a new subscription starts, the previous cycle's make-up opportunity is forfeited — no carry-over
- Make-up does NOT add extra sessions — it fills a previously absent slot
- Staff selects: student → original absent session → available compatible slot
- System shows only compatible slots: same or adjacent level, same branch

4.6 Freeze System

Freeze = Complete temporary suspension of all student activity. No automation or extension occurs.

Rule	Behavior
Visibility	Student completely hidden from all sessions while frozen
Attendance	Not tracked — no records created during freeze
Subscription	NOT extended — freeze simply pauses activity with no auto-resume
Payments	No payment required or triggered during freeze period
Data	100% preserved — nothing deleted or reset
Reactivation	Admin or Staff can reactivate the student at any time
Settings	Max freeze days per month configurable by Super Admin

Use cases: illness, travel, seasonal break, or long-term departure with possibility of return.

4.7 Coach Management, Check-in & Payroll

- Each coach has a login account with: Name, Phone, Email, Hourly Rate (set by Super Admin)
- Coach checks IN at session start and checks OUT at session end via web or mobile browser
- System auto-calculates hours worked = Check-out time minus Check-in time
- Multiple coaches per session supported — e.g., substitute takes over mid-session
- Monthly payroll auto-calculated: sum of all session hours x hourly rate (private sessions included)
- Staff cross-validates coach attendance — both entries visible to Admin for verification

Location-Based Check-in / Check-out Validation

To prevent remote check-ins, the system validates the coach GPS location against the academy registered coordinates before accepting any check-in or check-out.

Rule	Behavior
Academy coordinates	Latitude + Longitude stored in System Settings by Super Admin
Allowed radius	Configurable default 100 meters — set in System Settings
On check-in	Browser or mobile sends current GPS. System calculates distance via Haversine formula.
If within radius	Check-in recorded as VALID with timestamp and coordinates
If outside radius	Check-in recorded as INVALID — soft validation: entry saved but flagged. Admin sees alert. Coach NOT blocked.
Check-out	Same validation logic applied on check-out
Stored data	check_in_lat, check_in_lng, check_out_lat, check_out_lng, location_status (valid / invalid)

Soft validation is intentional — GPS drift and weak signals can cause false negatives. Admin reviews flagged entries rather than blocking the coach entirely.

4.7b Staff Login & Logout Tracking

Every staff login and logout is automatically recorded. This provides attendance visibility for management without requiring a physical fingerprint or badge system.

Event	Data Recorded	Purpose
Login	staff_id, login_time, login_lat, login_lng	Verify staff arrived on time
Logout	staff_id, logout_time, logout_lat, logout_lng	Verify end of shift
Hours calculated	logout_time minus login_time	Visibility into working hours per day
Location flag	valid / invalid (same radius logic as coaches)	Detect off-site logins

- Admin can view a daily log: who logged in, at what time, from where
- Late arrivals and early logouts are visible immediately on the Admin dashboard
- No blocking — all logins are accepted, flagged entries highlighted for Admin review

- Logout time auto-recorded when the session expires if staff forgets to log out manually

4.8 Student Class Transfers

Step	Actor	Action
1	Staff	Submits transfer request: Student, From Class, To Class, Reason
2	Admin	Reviews request — Approves or Rejects
3 (if Approved)	System	Removes student from old class, adds to new class
4	System	Logs transfer record for audit trail

4.9 Exam System & Level Promotion

- Admin creates exams: Name (Mid-Year / Final), Date, Fee, Target Level
- Students appear on the exam screen with full profile details
- Staff/Admin records result per student: Pass or Fail
- On PASS: system automatically promotes student to the next level (as configured in Settings)
- On FAIL: student remains at current level
- Exam fee is automatically logged as a payment transaction
- Level order and exam fees fully managed from the Super Admin Settings panel

4.10 Payment & Revenue Tracking

Every financial transaction is recorded as a Payment entry linked to the student.

Payment Type	العربي	Example
Subscription	اشتراك	4-session cycle fee
Product	مايوهات / منتج	Leotard ready-stock sale
Event	فعالية	Recital or TV show enrollment fee
Private Session	برايفيت	1-on-1 coaching fee
Exam Fee	رسوم امتحان	Level exam registration
New Enrollment	حجز جديد	First registration fee

- Payment methods: Cash or InstaPay
- Partial payments supported — remaining balance tracked and displayed
- Staff member who recorded the payment is logged with timestamp
- Daily revenue auto-calculated from all payment transactions

4.11 Daily Expense Tracking

- Staff records expenses: description, amount, date, hall/branch
- Examples: electricity, internet, cleaning supplies, gas, miscellaneous
- Net daily revenue = Total payments – Total expenses (shown on dashboard)

4.12 Leotard Inventory

- Each product: Name, Size, Cost Price, Selling Price, Stock Quantity, Type (Ready / Made-to-Order)
- Ready stock: quantity deducted from inventory automatically on each sale
- Made-to-Order: creates an order record with student, size, tailor cost, status (Pending / In Production / Done / Delivered)
- Tailor supplier linked to cost tracking

4.13 Events & Activities

- Admin creates events: Name, Type, Date, Venue, Price
- Types: Recital, TV Show, Workshop, Competition
- Students enrolled by staff — payment status tracked per student
- Event fees auto-logged to the payments module

4.14 Private Sessions

- Recorded per session: Student, Coach, Duration (hours), Date, Fee
- Fee auto-logged as a payment transaction
- Coach hours from private sessions included in monthly payroll calculation

4.15 Super Admin Settings Panel

Managed exclusively by the Super Admin. All key system parameters are controlled here.

Setting	Description
Subscription Plans	Sessions per cycle (default: 4), price per plan
Ballet Levels	Level names and promotion order (1, 2, 3...)
Exam Fees	Fee per exam type
Max Freeze Days	Maximum freeze days allowed per month per student
Absence Auto-hide	Number of consecutive absences before auto-Inactive
Hall Configuration	Hall 1 and Hall 2 names and capacity
User Management	Create / edit / deactivate staff and coach accounts

5. Revision Log

Version	Key Changes
v1.0	Initial draft — basic student, class, and attendance features only
v1.5	Added: events, revenue tracking, inventory, tailoring orders, private sessions
v2.0	Single branch (Miami) with Hall 1 & Hall 2. Freeze logic finalized. Make-up within same subscription only. Multi-coach per session. Transfer workflow. Exam auto-promotion. Partial payments. Absence auto-hide. Bilingual AR/EN. Settings panel.
v3.0 (Final)	Added: Location-based check-in/out for coaches (GPS + Haversine, soft validation, 100m radius configurable). Staff login/logout tracking with GPS and timestamp. location_status field (valid/invalid). Academy coordinates and allowed_radius in System Settings. Staff_Attendance table. New User Stories US-41 to US-46.